

Illinois Campaign Finance Intelligence Platform

Comprehensive 7-Phase Analysis

Product Engineering · UX · Growth · Data Science · Monetization

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Executive Summary

The Illinois Campaign Finance Intelligence Platform is a sophisticated civic-tech application tracking Illinois local and federal campaign finance data. Built with Python 3.12, Flask 3.0, and SQLite (WAL mode), it manages a 5.5 GB database with 38 tables spanning 6.4M+ state receipts and 20K+ federal contributions. The platform features 11 Flask blueprints, 40+ routes, 38 Jinja2 templates, Playwright-based web scraping, FEC API integration, and a materialized analytics engine with network analysis, anomaly detection, and donor entity resolution.

Key strengths include a deep data model, a mature analytics layer (HHI, Gini, centrality scoring), and strong entity resolution with confidence-scored merging. Critical gaps include missing CSRF protection, no API authentication or rate limiting, SQL injection risks in sort parameters, unbounded query results on several pages, and CSV exports that buffer 500K rows in memory. The platform has strong bones for scaling into a defensible civic-tech SaaS product.

Top 10 Quick Wins

#	Action	Category	Effort	Priority
1	Add CSRF protection	Security	2 hrs	High
2	Parameterize SQL sort fields	Security	1 hr	Critical
3	Add streaming CSV exports	Performance	3 hrs	High
4	Paginate report contributions	Performance	2 hrs	High
5	Set production SECRET_KEY from env	Security	15 min	Critical
6	Add API rate limiting	Security	2 hrs	Medium
7	Add connection pooling	Performance	2 hrs	Medium
8	Add login rate limiting	Security	1 hr	Medium
9	Add 500 error handler	UX	30 min	Low
10	Add cache-busting versioned static assets	UX	1 hr	Low

NOW / NEXT / LATER Roadmap

Horizon	Key Deliverables
NOW (0-4 wks)	CSRF + SQL injection fix Streaming CSV Paginate all unbounded queries API auth + rate limiting Production secret key
NEXT (1-3 mo)	Full-text search (FTS5) Contribution velocity scoring Donor similarity clusters Saved filter presets + shareable URLs Mobile-responsive tables

Horizon	Key Deliverables
LATER (3-6 mo)	Graph export (CSV, GraphML) Predictive donor modeling Public API with tiered access Email/Slack alerts for anomalies Federal/state cross-network explorer

Phase 0 — Repo Recon

Stack & Architecture

Component	Details
Language	Python 3.12
Web Framework	Flask 3.0 with 11 Blueprints
Database	SQLite 5.5 GB, WAL mode, 38 tables
Scraping	Playwright (async), ASP.NET postback handling
CLI	Click command groups
Templates	Jinja2 (38 templates), raw SVG charts
API Integration	FEC API with caching via raw_extractions
Deployment	Hetzner VPS, gunicorn, systemd timers
Testing	pytest + pytest-asyncio, 12 modules (~5,300 lines)

Directory Structure

Directory	Contents
cli/	Click commands: scraping, FEC sync, bulk import, maintenance, analytics refresh
database/	Schema, models, analytics engine (2,265 lines), bulk loader, FEC client, entity resolution
webapp/	Flask app factory, 11 route modules, 38 templates, 3 JS files, CSS
tests/	12 test modules covering parsing, analytics, FEC, entity resolution, webapp routes
docs/	Feature checklist, bulk mapping, data expansion roadmap, update guide
data/	SQLite DB file, bulk download staging

Data Model Summary

Layer	Tables
Core (SBE)	committees, reports, donors, contributions, raw_extractions, scrape_state
D-2 Reports	d2_reports, d2_itemized_links, d2_itemized_entries
Bulk Download	bulk_committees_clean, bulk_d2_totals_clean, bulk_candidates_clean, bulk_receipts_clean + 7 join/agg tables
FEC Federal	fec_il_candidate_seed, fec_candidate_match, fec_candidate_committees, fec_schedule_a_contributions, fec_local_donors
Analytics	analytics_donor_committee_agg, analytics_committee_monthly_totals, analytics_large_contributions, analytics_donor_summary
Entity Resolution	donor_entity_local, donor_entity_local_member
Admin	app_users, manual_entry_queue

Quality & Security Gaps

- SQL injection risk in models.py sort parameter (f-string interpolation without whitelist)
- No CSRF protection on form routes (Flask-WTF not used)
- No API authentication or rate limiting on 15 JSON endpoints
- Production SECRET_KEY defaults to 'dev-secret-key-change-in-production'
- CSV exports buffer up to 500K rows in memory (no streaming)
- Report detail page loads ALL contributions without pagination
- No connection pooling — each request opens a new SQLite connection
- No 500 error handler configured
- Scraper commands (Playwright automation) have 0% test coverage
- No full-text search — large text columns unindexed

Phase 1 — High-Impact Improvements

15 Shippable Improvements

#	Improvement	Category	Priority	Details
1	Add CSRF tokens to all forms	Security	High	All POST routes lack CSRF. Add Flask-WTF or manual token.
2	Whitelist SQL sort fields	Security	Critical	models.py uses f-string for ORDER BY. Validate against allowed list.
3	Stream CSV exports	Performance	High	Replace in-memory buffering with generator + StreamingResponse.
4	Paginate report contributions	Performance	High	/reports/<id> loads all rows. Add LIMIT/OFFSET pagination.
5	Set SECRET_KEY from env	Security	Critical	Hardcoded dev key. Read from FLASK_SECRET_KEY env var.
6	Add API rate limiting	Security	Medium	flask-limiter with per-IP throttling on /api/ endpoints.
7	Add connection pooling	Performance	Medium	Each get_db() opens new connection. Use connection pool.
8	Add login rate limiting	Security	Medium	No brute-force protection on /auth/login. Add delay/lockout.
9	Add 500 error handler	UX	Low	Unhandled exceptions show raw traceback. Add friendly error page.
10	Pagination on committee contributions	UX	Medium	Hard-limited to 100 rows with no 'next page'. Add pagination.
11	Add FTS5 full-text search	Feature	Medium	Text searches scan full tables. SQLite FTS5 for descriptions.
12	Atomic materialization refresh	Data	Medium	Partial refresh leaves stale data. Wrap in transaction.
13	Snapshot build progress indicator	UX	Low	Full-mode analytics shows no progress. Add polling endpoint.
14	API response envelope	API	Low	Add consistent {data, meta, errors} JSON envelope.
15	Consolidate donor detail routes	UX	Low	3 routes (id/key/entity) confuse users. Unify with smart routing.

10 Paper Cuts

#	Paper Cut	Fix
1	Search results show no total count	Add 'Showing X of Y results' header
2	Compare mode loses selections on dropdown change	Debounce form submission
3	No 'back to list' breadcrumb on detail pages	Add consistent breadcrumb nav
4	Date filters use plain text input on some pages	Standardize HTML5 date inputs
5	Truncated names have no tooltip	Add title attribute for full text on hover
6	Flash messages don't auto-dismiss	Add JS fade-out after 5 seconds
7	Tables not responsive on mobile	Add horizontal scroll wrapper
8	No empty-state illustrations	Add friendly messages when no data found
9	Sort direction indicators inconsistent	Standardize across all sortable tables
10	Footer disclaimer is dense	Collapse into expandable accordion

Phase 2 — Feature Ideation & Prioritization

Core Civic-Tech Features (10)

#	Feature	Description
1	Election Countdown Dashboard	Real-time days-to-election with fundraising pace tracker
2	Candidate Report Card	Auto-generated score card: diversity, small-dollar %, transparency
3	Voter-Facing Contribution Lookup	'Who funds my candidates?' search by address/district
4	Dark Money Tracker	Flag committees with opaque funding sources or shell donors
5	Legislative Vote + Money Correlation	Cross-reference voting records with top donors
6	Lobbyist-to-Contribution Pipeline	Match registered lobbyists to contribution patterns
7	Public Campaign Finance Scorecard API	Open API for journalists and researchers
8	Civic Engagement Alerts	Email/SMS when new large contributions filed in user's district
9	Historical Trend Explorer	Multi-cycle trend visualization with inflation adjustment
10	Redistricting Impact Analyzer	Show how district changes affect donor networks

Data Science Features (10)

#	Feature	Description
1	Contribution Velocity Scoring	Rate-of-change analysis flagging sudden fundraising spikes
2	Donor Similarity Clusters	K-means/DBSCAN grouping donors by giving patterns
3	Anomaly Detection ML	Isolation forest or autoencoder for unusual contribution patterns
4	Predictive Fundraising Model	Forecast next-quarter totals from historical trends
5	Network Community Detection	Louvain/Label Propagation to find donor communities
6	Influence Propagation Score	PageRank-style scoring across donor-committee-candidate graph
7	Donor Lifetime Value Model	Predict donor retention and future contribution amounts
8	Geographic Hotspot Analysis	Spatial clustering of contribution density by ZIP code
9	Text Classification for Expenditures	NLP model to categorize spending beyond keyword matching
10	Cross-Network Bridge Detection	Identify donors bridging otherwise disconnected networks

Non-DS Enhancements (10)

#	Feature	Description
1	Saved Filter Presets	Save and share complex filter combinations via URL
2	Shareable Report Links	Permalink to any filtered view with OG meta tags
3	Email Digest Subscriptions	Weekly summary of new filings in tracked races
4	Embeddable Widgets	iframe-ready charts for journalists and bloggers

#	Feature	Description
5	Bulk Data Export Portal	Self-service download of cleaned datasets (CSV/JSON)
6	Audit Log & Activity Feed	Track admin actions and data changes
7	Multi-State Expansion Framework	Abstract state-specific logic for other states' data
8	Dark Mode	CSS custom properties for theme switching
9	Accessibility Audit + Fixes	WCAG 2.1 AA compliance across all pages
10	Progressive Web App (PWA)	Offline support for saved reports and bookmarks

Phase 3 — Data Science Roadmap

10 Practical DS/Analytics Features

#	Feature	Approach	Horizon	Complexity
1	Contribution Velocity Scoring	Calculate 7/30/90-day contribution rate per committee. Flag acceleration > 2σ. Uses existing analytics_committee_monthly_totals.	Now	Low
2	Donor Similarity Clusters	K-means on (total_amount, frequency, committee_connection). Already prototyped in federal_fec.py.	Now	Low
3	Anomaly Detection ML	Isolation forest on contribution vectors. Upgrade from current rule-based flags in analytics.py.	Next	Medium
4	Network Community Detection	Louvain algorithm on donor-committee bipartite graph. Averages existing get_network_graph().	Next	Medium
5	Influence Propagation Score	PageRank on weighted donor→committee→candidate edges. Extends existing centrality scoring.	Next	Medium
6	Predictive Fundraising Model	Linear regression → gradient boosting on historical quarterly totals. Monthly retraining.	Later	High
7	Donor Lifetime Value	Survival analysis (Kaplan-Meier) on donor retention. Segmented by first contribution size.	Later	High
8	Geographic Hotspot Analysis	DBSCAN on ZIP centroids weighted by contribution density. Extends get_geo_summary().	Later	Medium
9	Expenditure Text Classification	Fine-tuned classifier replacing keyword NLP in get_np_text_summary(). 8→20+ categories.	After	High
10	Cross-Network Bridge Detection	Betweenness centrality on multi-component graph. Identify donors connecting partisan clusters.	Later	Medium

Easy Wins (ship in < 1 week)

- **Contribution Velocity Scoring** — Pure SQL over analytics_committee_monthly_totals. Add velocity_score column, flag committees with 30-day rate > 2× trailing average. Display as sparkline on candidate pages.
- **Donor Similarity Clusters** — Reuse existing K-means code from federal_fec.py. Run on analytics_donor_summary table. Store cluster_id in new column. Show cluster labels on donor list page.

Defensibility Features

- **Entity Resolution Quality Score** — Confidence-weighted merge quality metric. Surfaces low-confidence merges for human review. Builds on donor_entity_local scoring. Differentiator: most civic-tech tools skip entity resolution entirely.
- **Cross-Jurisdiction Network Analysis** — Federal↔state donor overlap network. Uses fec_local_donor_matches + analytics_donor_committee_agg. Shows money flowing across jurisdictional boundaries. Unique dataset advantage.

Data Quality Scoring System

A 0-100 composite score per entity measuring data completeness and reliability:

Dimension	Weight	Metric
Completeness	25%	% of non-null fields (name, address, occupation, employer, dates)
Freshness	20%	Days since last update vs. expected refresh cadence
Consistency	20%	Cross-table agreement (D-2 totals vs. receipt sums, entity merge confidence)
Coverage	20%	% of expected filings present (election cycles, quarterly reports)
Accuracy	15%	Reconciliation delta, duplicate rate, garbage detection score

Phase 4 — Monetization

#	Model	Description	Pricing	Status
1	Freemium Civic-Tech SaaS	Free public dashboard; paid tiers for API, alerts, bulk CSV export, custom reports	\$0, \$29, \$99, custom	Recommended
2	API Access Tiers	Free: 100 req/day; Pro: 10K req/day; Enterprise: unlimited	\$0, \$29, \$99, custom	Recommended
3	Consulting & Custom Analysis	On-demand research reports for campaigns, journalists, NGOs	\$500-\$1,000/report	Supplement
4	Data Licensing	Cleaned, linked datasets for academic research, NGOs	\$1,000-\$10,000/org	Supplement
5	Sponsored Research	Foundation/nonprofit-funded investigations published as open reports	Grant-funded	Long-term
6	Training & Workshops	Campaign finance data literacy for journalists, NGOs	\$200-\$1,000/session	Long-term

Recommended Pricing Tiers

Tier	Includes	Price
Free (Civic)	Full public dashboard, search, compare, basic analytics	\$0
Pro (Researcher)	API access (10K req/day), CSV bulk export, saved filters, email alerts	\$29/mo
Team (Newsroom)	Everything in Pro + 5 seats, priority data freshness, custom reports	\$99/mo
Enterprise	Unlimited API, webhooks, white-label embeds, SLA, dedicated support	Custom

Paywall Strategy

- Keep all current public-facing pages free forever — civic transparency is the mission.
- Gate power-user features: API access, bulk CSV export (>1,000 rows), saved filter presets, email alerts.
- Gate data freshness: free tier gets weekly refresh; paid gets daily + real-time FEC sync.
- Gate depth: free shows top-50 donors; paid shows full donor list with entity resolution details.
- Never gate: basic search, candidate lookup, contribution totals, anomaly flags.

Phase 5 — Growth & Promotion

10 Distribution Channels

#	Channel	Strategy
1	Civic Tech Twitter/X	Share visualizations, anomaly flags, election-cycle insights
2	Reddit r/illinois, r/politics, r/dataisbeautiful	Post data visualizations with source links
3	Hacker News	Show HN posts on technical architecture, entity resolution, data pipeline
4	Local Journalism Partnerships	Provide data to reporters covering IL campaigns
5	University Data Science Programs	Offer as teaching dataset for Northwestern, UChicago, UIUC
6	Civic Tech Meetups	Chicago civic tech, Code for America brigade presentations
7	Substack / Blog	Weekly 'Follow the Money' analysis posts with embedded charts
8	LinkedIn	Build-in-public technical content (see Phase 6)
9	Government Transparency Orgs	Partner with Sunlight Foundation, OpenSecrets, BGA
10	SEO / Content Marketing	Target 'Illinois campaign finance' + candidate name long-tail queries

5 Partnership Targets

#	Partner	Value Proposition
1	Better Government Association (BGA)	Chicago-based watchdog; could co-publish investigations
2	Illinois Campaign for Political Reform	Advocacy org; could embed our data in their reports
3	Block Club Chicago / Chicago Sun-Times	Newsrooms covering local politics; data partnership
4	Northwestern Medill Data Journalism	Student projects using our API; credibility + backlinks
5	OpenSecrets / FollowTheMoney.org	Cross-link federal/state data; mutual SEO benefit

5 Virality Loops

#	Loop	Mechanism
1	Shareable Candidate Cards	Auto-generated OG-image cards for any candidate (link → share → traffic)
2	Embeddable Charts	Copy-paste iframe widgets for journalists writing about IL races
3	'Check Your District' Widget	Enter address → see who funds your representatives → share result
4	Anomaly Alerts on Social	Auto-post when large/unusual contributions detected
5	Data Quality Leaderboard	Gamify completeness: 'Help us verify 100 more contributions'

30-Day Launch Plan

Week	Actions
Week 1	Ship CSRF + SQL injection fixes. Write 'Why I Built This' blog post. Post to HN Show HN.
Week 2	Add streaming CSV + paginated reports. Reach out to BGA and 2 journalists. Post Reddit r/dataisbeautiful visualization.
Week 3	Launch shareable candidate cards. Submit to civic tech directories. Present at Chi Hack Night.
Week 4	Publish API documentation. Contact Northwestern Medill. Post LinkedIn case study. Measure: traffic, API signups, press mentions.

Phase 6 — LinkedIn Content Plan

10 LinkedIn Post Ideas

#	Post Title	Angle
1	I tracked \$2B in IL campaign money. Here's what the data reveals.	Hook: scale + discovery
2	How I built entity resolution for 6.4M political donations (no ML required)	Technical credibility
3	The 5 patterns that predict campaign finance anomalies	Insight-led, shareable
4	Why SQLite handles our 5.5 GB political finance database just fine	Contrarian tech take
5	I found the donors funding both sides of IL's biggest races	Controversy + data
6	Building a civic tech product alone: lessons from 18 months of scraping	Founder story
7	The entity resolution problem nobody talks about in government data	Educational, niche authority
8	How federal and state campaign money connects (and why it matters)	Cross-jurisdiction insight
9	I automated Illinois campaign finance monitoring. Here's the architecture	System design content
10	Open data isn't enough. Here's what it takes to make it useful.	Thought leadership

2 Build-in-Public Series

Series 1: 'Scraping Government Websites' (5 parts)

- Part 1: Why government data is stuck in the 1990s
- Part 2: Reverse-engineering ASP.NET postbacks with Playwright
- Part 3: Building a resilient scraper (rate limiting, retry, resumability)
- Part 4: Parsing messy financial tables at scale
- Part 5: From raw HTML to queryable analytics

Series 2: 'Entity Resolution for Civic Data' (4 parts)

- Part 1: The donor deduplication problem (why 'John Smith' isn't one person)
- Part 2: Jaccard similarity, confidence scoring, and bridge rules
- Part 3: Building a human-in-the-loop merge review system
- Part 4: Matching donors across federal and state systems

2 Case Studies

- **Case Study 1:** 'How we identified \$X in potentially misreported contributions using D-2 reconciliation' — Walk through the bulk_d2_receipts_recon pipeline, show real discrepancy examples, explain the audit flag system.
- **Case Study 2:** 'Mapping Illinois's donor network: 6.4M contributions, one graph' — Visualize the network analysis output, highlight power players, show community detection results, demonstrate the civic value.

1 Technical Deep Dive

'Materialized Analytics at Scale with SQLite' — How we serve complex analytics dashboards from a single SQLite database. Covers: WAL mode tuning, materialized view refresh strategy, snapshot caching with SHA1(params) keys, background ThreadPoolExecutor refresh, HHI/Gini computation, and why this architecture works up to ~10 GB before needing PostgreSQL.

Phase 7 — Risk Register

ID	Risk	Severity	Likelihood	Category	Mitigation
R1	SQL Injection	Critical	High	Security	Whitelist sort fields; parameterize all dynamic SQL
R2	Missing CSRF	High	High	Security	Add Flask-WTF CSRF tokens to all forms
R3	No API Auth	High	High	Security	Add API key auth + rate limiting via flask-limiter
R4	Hardcoded Secret Key	Critical	High	Security	Move to env var; rotate immediately in production
R5	Memory OOM on CSV Export	High	Medium	Performance	Implement streaming response with generator
R6	Unbounded Report Query	High	Medium	Performance	Add pagination to /reports/<id> and /api/reports/<id>
R7	ISBE Website Changes	Medium	High	Data	Add HTML structure validation tests; alert on parse failures
R8	FEC API Rate Limits	Medium	Medium	Data	Implement proper exponential backoff; cache aggressively
R9	Stale Analytics Cache	Medium	Medium	Data	Add cache invalidation on data refresh; show freshness badge
R10	Single-Server Deployment	Medium	Low	Infra	Document recovery procedure; add automated backups
R11	No Audit Trail	Medium	Low	Compliance	Log all admin actions; add activity feed
R12	Privacy (Donor PII)	High	Low	Legal	All data is public record; add disclaimer; no SSN/DOB stored
R13	Bus Factor = 1	High	Medium	Maintainability	Document architecture; add contributor guide; write ADRs
R14	Test Coverage Gaps	Medium	Medium	Quality	Add scraper tests with mocked HTML; increase to 80% coverage
R15	SQLite Scale Ceiling	Low	Low	Scalability	Monitor DB size; plan PostgreSQL migration path at ~10 GB

Implementation Notes — Top 5 Roadmap Items

1. CSRF Protection

- **File:** webapp/app.py — Add Flask-WTF CSRFProtect(app) in create_app()
- **Files:** All 38 templates with <form> tags — Add {{ csrf_token() }} hidden input
- **Dependency:** pip install flask-wtf
- **Test:** Verify POST requests without token return 400

2. SQL Sort Field Whitelist

- **File:** database/models.py — Committee.get_all(), Report.get_all(), Donor.get_all()
- **Fix:** Replace f-string ORDER BY with ALLOWED_SORTS dict lookup
- **Pattern:** sort_field = ALLOWED_SORTS.get(sort_field, 'id')
- **Test:** Verify injection attempt ('id; DROP TABLE') is rejected

3. Streaming CSV Export

- **Files:** webapp/routes/candidate_finance.py, d2_receipts_recon.py
- **Fix:** Replace list accumulation with generator function
- **Pattern:** return Response(stream_with_context(generate()), mimetype='text/csv')
- **Impact:** Memory usage drops from O(n) to O(1) for 500K-row exports

4. Report Pagination

- **File:** webapp/routes/reports.py — report_detail() function
- **File:** webapp/routes/api.py — api_report_detail() function
- **Fix:** Add page/per_page params with LIMIT/OFFSET query
- **Template:** webapp/templates/reports/detail.html — Add pagination controls

5. API Rate Limiting

- **File:** webapp/app.py — Add flask-limiter initialization
- **File:** webapp/routes/api.py — Add @limiter.limit() decorators
- **Config:** Default 100/hour for anonymous; 1000/hour for API key holders
- **Dependency:** pip install flask-limiter